WHAT SURVIVES THE ETH REFUTATIONS? A MAP OF FINE-GRAINED HYPOTHESES AND CONDITIONAL LOWER BOUNDS

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Abstract. Many lower bounds in exact, parameterised and fine-grained complexity are proved from the Exponential Time Hypothesis (ETH) or the Strong ETH. In October 2026 OpenAI and Anthropic claimed deterministic subexponential-time algorithms for #k-SAT that refute both. However, a refuted hypothesis does not settle the theorems proved from it: what falls depends on the premise that each proof actually uses, and we are unaware of a systematic account. We present a map of 33 hypotheses and 27 published conditional theorems, assuming the claimed algorithms are correct. Each row is refuted, has lost its premise, or survives; a fourth label marks the few theorems whose conclusion is now contradicted by an explicit algorithm. The map rests on one distinction. The new algorithms handle k-CNF formulas of bounded width and, through the width reduction of Calabro, Impagliazzo and Paturi, CNFs of any width with at most $n \cdot 2^{o((\log n)^{1/4}/\sqrt{\log\log n})}$ clauses (a short derivation of ours from the explicit rate of the Anthropic manuscript). Nothing is known for CNF-SAT with polynomially many clauses of unbounded width. Many of the best-known lower bounds (treewidth optimality, edit distance, Fréchet distance, LCS, diameter approximation) are proved from that "CNF-SETH" and survive with their premise intact. Gap-ETH and the ETH-based proof of the Parameterized Inapproximability Hypothesis lose their force; Almost-ETH, OV, 3SUM, APSP and the Set Cover Conjecture are untouched. We list the few new algorithms that follow. A refutation removes premises; it does not by itself give algorithms, and nothing here is practical. Everything is conditional on the claimed refutations, which are under review. The supporting notes are in our public repository, https://github.com/xennygrimmato/eth-hamiltonicity.

1. Introduction

A large part of exact, parameterised and fine-grained complexity rests on two hypotheses. The Exponential Time Hypothesis (ETH) of Impagliazzo and Paturi [39, 40] defines s_k, for $k \geq 3$, as the infimum of the δ such that k-SAT on n variables is solvable in time $O(2^{\delta n})$, and asserts $s_k > 0$ for $k \geq 3$. The Strong ETH (SETH) is $s_\infty = \lim_k s_k = 1$ [40]. Lower bounds proved from them range from $2^{o(n)}$ bounds for NP-hard graph problems [41, 49] to quadratic bounds for problems in P, such as edit distance [7].

1.1. The two refutations.

In October 2026 two manuscripts appeared, by OpenAI [57] and by Anthropic [22]. Each claims a deterministic algorithm that counts the satisfying assignments of a k-CNF in subexponential time. Both come with Lean 4 formalisations of their top-level statements. These are *claimed* results, still under review. We write [A] and [O] for them, and use them in the following forms.

Imported Theorem 1.1 ([A], Main Theorem, eq. (1))**.** *There is an absolute constant C and a deterministic algorithm that counts the models of a k-CNF with n variables and m clauses in time*

$$2^{C\cdot\min\{k^2 n \log\log n/\sqrt{\log n},\ km/\sqrt{\log m}\}}.$$

Imported Theorem 1.2 ([O], main theorem)**.** *For each fixed k there is a deterministic algorithm that counts the models of a k-CNF with n variables and m clauses in $(n+m)^{O_k(1)}2^{O_k(n\log\log n/\sqrt{\log n})}$ bit operations and polynomial space, and in $2^{O_{k,D}(n/\sqrt{\log n})}$ when $m \leq Dn$.*

Two features matter below. First, Imported Theorem 1.1 is *uniform in k*: C is absolute, so the bound is subexponential in n as long as $k = o((\log n)^{1/4}/\sqrt{\log\log n})$. [O] states no dependence on k, so it is not known to be uniform. Uniformity is irrelevant for ETH and SETH. Each engine gives $s_k = 0$ for every fixed

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k, hence $s_\infty = 0$ ([A], Corollary 5; the reduction from 3-SAT to k-SAT is the deterministic sparsification of [41]). It matters only for formulas whose width grows with n (Section 2). Second, both algorithms are deterministic and count. So they refute the randomized, counting and nondeterministic variants of ETH and SETH *for k-CNFs* at once ([A], Corollaries 5 and 6 and the remark after Corollary 5).

1.2. **The problem: which results fall?** However, a refutation of ETH does not tell us which of the results proved from it fall. A conditional lower bound "$H \Rightarrow$ problem X needs time T" is proved by a reduction *from* a problem covered by H *to* X. Refuting H removes the premise; it says nothing about X. Worse, the premise named in a theorem is often stronger than the premise its proof uses. We are unaware of a systematic account of what the refutations affect, and without one every result stated "under ETH" or "under SETH" is in doubt.

We consider the *classification problem*: for each fine-grained hypothesis and each published theorem $H \Rightarrow X$, decide what follows from [A] and [O]. We take the view that the status of a theorem depends on its premise *as actually used in the proof*, not as stated. Throughout, "refuted" means *refuted assuming the claimed refutations*, and we use the following statuses.

- **refuted**: the hypothesis itself is false.
- **premise refuted**: a theorem $H \Rightarrow X$ with H false; nothing here contradicts X.
- **survives**: we know of no consequence of the refutations that touches the statement. For a theorem this means that its premise, as actually used in the proof, still stands.

The additional label **conclusion refuted** marks a theorem $H \Rightarrow X$ whose conclusion X is contradicted by an explicit algorithm. One row is a theorem whose premise is the *negation* of ETH, so that the theorem now **fires** (Section 4.1). A few rows are **mixed** and are split in the reason column.

Example 1.3 (Running example)**.** Lokshtanov, Marx and Saurabh [48] prove that Independent Set has no $(2-\varepsilon)^{\mathrm{tw}}$ algorithm, under SETH. By the classical argument [41, 49], ETH implies that Independent Set on n vertices has no $2^{o(n)}$ algorithm. Both theorems cite a refuted hypothesis, yet they end up in different places. The second has its conclusion refuted: [A], Corollary 7, gives $O(2^{\delta n})$ for every $\delta > 0$ (row B13). The first survives: its proof reduces from CNF-SAT with m clauses of unbounded width, which the new algorithms do not reach (row B1, Section 2). We return to this pair in Section 6.

1.3. **The naive route and the principle.** At first one might expect every SETH-based bound to fall with SETH, and every ETH-based bound with ETH. However, Example 1.3 shows that this is wrong in both directions. A bound can survive because its proof never needs bounded width, and a refutation can contradict a conclusion only through an algorithm, never through the reduction that proved the bound. Instead, we rely on two rules. A row falls only if the premise its proof uses falls. And a row gains an algorithm only if some reduction runs in the other direction: from X to k-SAT with a linear number of variables (the SERF framework of [41], and [A], Corollary 7), or from more general theorems inside the manuscripts that the top-level statements do not expose (our own sparse results, Section 5). Most rows therefore lose a premise and gain nothing. We say so row by row.

1.4. **Contributions.** Our contributions are:

- We separate bounded from unbounded width. We prove that Imported Theorem 1.1 counts the models of CNFs of any width with at most $n \cdot 2^{o(L(n))}$ clauses in time $2^{o(n)}$, where $L(n) = (\log n)^{1/4}/\sqrt{\log\log n}$, and that nothing follows at $n^{1+\Omega(1)}$ clauses (Section 2).
- We classify 33 hypotheses: 12 are refuted, 20 survive and 1 is mixed (Section 3).
- We classify 27 conditional theorems: 4 have their conclusion refuted, 1 fires, 10 lose their premise, 9 survive and 3 are mixed (Section 4).
- We list the new algorithms that follow, and separate those that need only the formalised top-level theorems from those that need internal results of [A] (Section 5).
- We read the map by area and relate it to the literature (Sections 6 and 7).

Much of the classification is bookkeeping that an expert in each area could do. What we add is one map across the areas, with each row checked against its primary source where one exists.

1.5. **Sources and status of each claim.** The map was compiled from the primary sources. For every row tagged [L] in Tables 1 and 2 we checked the wording against the abstract or full text of the cited source (arXiv, ECCC, DROPS, journal, or authors' copy). Rows tagged [F] are folklore conjectures without a single primary statement; [I] marks our own inference, and [O] our own results. In an earlier pass, nine rows rested on secondary sources or on sources we had not re-read (ETH itself; the CIP duality; Williams' Max-2-CSP algorithm; Feige's hypothesis; Pătraşcu–Williams, twice; Cai–Juedes; Chen–Huang–Kanj–Xia; Marx's treewidth paper). We re-verified all nine against the primary text or abstract for this note, so no row now rests on a secondary source. Theorem numbers refer to the versions named in the bibliography.

The working map behind this note, `research/fine-grained-map.md`, names for each row the source we checked, with its URL where there is one. It is in our public repository `https://github.com/xennygrimmato/eth-hamiltonicity` [68], together with the Lean proofs of our own results that we cite. We separate three kinds of claim. *Proved on paper*: Proposition 2.1 and the observations of Section 5. *Machine-checked in Lean*: only the results of ours that Section 5 labels so; each uses only the standard axioms and is built on the labs' own Lean developments. *Conjectured, not claimed*: the further sparse problems of Section 5. Everything rests on [A] and [O].

Outline. Section 2 separates bounded from unbounded width. Sections 3 and 4 give the two tables. Section 5 lists the new algorithms, Section 6 reads the map by area, and Sections 7 to 9 give related work, a summary and the limitations.

2. Bounded width falls, unbounded width stands

In this section we state three facts used throughout, derive the density frontier of the new algorithms, and name the hypothesis that survives.

2.1. **Three facts.**

(F1) $s_k = 0$ for every k. This refutes ETH, SETH, #ETH, #SETH, $\oplus$SETH and NSETH in their k-CNF forms.

(F2) *Local witnesses.* A problem with complexity parameter n whose yes-instances have an $O(n)$-bit witness checkable by an $O(n)$-gate circuit, or by bounded-width clauses on $O(n)$ variables, gets deterministic time $2^{\delta n}$ for every $\delta > 0$ by Tseitin's transformation. Counts follow when the encoding is parsimonious. This is the SERF route [41]; [A], Corollary 7, uses it with a cardinality circuit of $O(n)$ gates.

(F3) *The width frontier* (Proposition 2.1 below). CNFs of arbitrary width are reached only when the number of clauses is barely superlinear.

2.2. **The width reduction and the density frontier.** Calabro, Impagliazzo and Paturi [16] relate CNF-SAT of bounded density to k-SAT by Schuler's width reduction. While some clause C has more than k literals, pick k of them, C', and branch on "C' is satisfied" (replace C by C') versus "C' is falsified" (set its k literals to false). The branches are disjoint, so model counts add up, and the leaves are k-CNFs. This counting form is [67, Lemma 6.1]. With constant k it gives #CNF-SAT with $m \le cn$ clauses in time $2^{\varepsilon n}$ for all $c, \varepsilon > 0$, from either engine [67, Corollary 6.2]. The uniformity of Imported Theorem 1.1 lets k grow, and gives a little more.

Proposition 2.1 (our derivation from Imported Theorem 1.1)**.** *Put $L(n) = (\log n)^{1/4}/\sqrt{\log\log n}$. Assume Imported Theorem 1.1. For every function f with $f(n) = o(L(n))$ there is a deterministic algorithm that counts the models of CNF formulas of arbitrary width with n variables and at most $n \cdot 2^{f(n)}$ clauses in time $2^{o(n)}$.*

Proof. Let $m \le Dn$ with $\log D = f(n)$, and fix $k \ge 2$, to be chosen. Run the width reduction. Along a root-to-leaf path every "satisfied" branch removes one long clause, so there are at most m of them, and every "falsified" branch fixes k variables, so there are $r \le n/k$ of them. The leaves with exactly r falsified branches number at most $\binom{m+r}{r}$ [16, proof of Lemma 5], so there are at most $(n/k+1)\binom{m+\lfloor n/k \rfloor}{\lfloor n/k \rfloor} \le (n+1)\big(e(Dk+1)\big)^{n/k}$ leaves. Each leaf is a k-CNF with $n' \le n$ variables and at most m clauses. If $n' \le \sqrt{n}$, count by brute force in time $2^{\sqrt{n}}\,\mathrm{poly}(m)$. Otherwise $\log n' \ge \frac{1}{2}\log n$, so $\log\log n'/\sqrt{\log n'} \le 2\log\log n/\sqrt{\log n}$ for n large, and Imported Theorem 1.1 counts the leaf in time 2^{sn} with $s = 2Ck^2 \log\log n/\sqrt{\log n}$. The total time is at most

$$\mathrm{poly}(n,m) \cdot 2^{\,(n/k)\,(f(n)+\log(k+1)+2)\,+\,sn\,+\,\sqrt{n}}.$$

Take $k = \lceil \sqrt{\max\{f(n), 1\} \cdot L(n)} \rceil$. Then $k \to \infty$, $f(n)/k \le \sqrt{f(n)/L(n)} \to 0$ and $(\log k)/k \to 0$, while $k/L(n) \to 0$ gives $s = 2C(k/L(n))^2 \to 0$. Every term of the exponent is $o(n)$, and $\mathrm{poly}(n, m) = 2^{o(n)}$ since $m \le n^{1+o(1)}$. □

Remark 2.2 (what does not follow).

(a) The clause-count branch of Imported Theorem 1.1 does worse here: it needs $km = o(n\sqrt{\log m})$, which allows only $m \le n(\log n)^{1/2-o(1)}$.

(b) Through [O] only constant density is reached explicitly, since its O_k-constants are not stated; a diagonal argument gives some unbounded but non-explicit density.

(c) For $m = n^{1+\Omega(1)}$, or $m \ge n \cdot 2^{(\log n)^{1/4+\Omega(1)}}$, nothing follows: the leaf count forces $k \gtrsim \log(m/n)$, where neither engine is subexponential in n. Hence the statement

CNF-SETH: for every $\varepsilon > 0$, CNF-SAT with n variables and m clauses of unbounded width is not solvable in time $2^{(1-\varepsilon)n}\,\mathrm{poly}(m)$

survives, together with its counting, parity and nondeterministic forms.

CNF-SETH is exactly the premise that many famous reductions use. Lokshtanov, Marx and Saurabh [48] assume that "SAT cannot be solved in $(2-\varepsilon)^n m^{O(1)}$ time" and stress that they need no sparsification. Backurs and Indyk [7] reduce from CNF-SAT with M clauses to edit distance. Bringmann's "SETH′" [12] is "no $O^*((2-\delta)^N)$ algorithm for CNF-SAT". Roditty and Vassilevska Williams [59, Theorem 1] conclude "CNF-SAT on n variables is in $O^*((2-\delta)^n)$". And the moderate-dimension Orthogonal Vectors conjecture follows from CNF-SETH by Williams' split-and-list reduction with $d = m$. What dies is everything that genuinely goes through sparsification to $O(n)$ clauses, or through bounded width.

3. Hypotheses: twelve refuted, twenty survive

In this section we classify the hypotheses. Table 1 lists them. "Gives algorithms" is noted in the reason column when a refutation also yields an explicit faster algorithm for the problem named.

Table 1: Hypotheses, assuming the claimed refutations.

	Hypothesis (source)	Status	Reason
A1	ETH: $s_k > 0$ for $k \ge 3$ [39, 40]^{L}	refuted	Either engine; 3-SAT and k-SAT in $2^{o(n)}$.
A2	SETH: $s_\infty = 1$ [40]; randomized algorithms allowed in [25]^{L}	refuted	(F1); [O] suffices despite non-uniformity in k.
A3	Density-SETH: CNF of any width with $\le cn$ clauses needs $2^{(1-\varepsilon)n}$ [16]; [55, Conj. 2]^{L}	refuted	Counting width reduction [67, Cor. 6.2].
A4	CNF-SETH: CNF-SAT $\notin 2^{(1-\varepsilon)n}\,\mathrm{poly}(m)$, unbounded width [48, 7, 12, 59]^{L}	survives	Proposition 2.1 and Remark 2.2: reach ends at $n2^{o(L(n))}$ clauses.
A5	Super-SETH: for every unbounded f, k-SAT $\notin 2^{n(1-f(k)/k)}$ [69]^{L}	refuted	It implies SETH.
A6	#ETH [30, p. 21:5]^{L}	refuted	[A], Corollary 6; [O]. Gives #3-SAT in $2^{o(n)}$.
A7	#SETH, ⊕SETH for k-CNF; equivalent to k-⊕Hitting Set and k-⊕Set Cover hardness [25, Thm. 1.3]^{L}	refuted	(F1).
A8	#CNF-SETH, ⊕CNF-SETH (poly(m) clauses)^{I}	survives	As A4.
A9	NSETH: $\forall\varepsilon\,\exists k$, k-TAUT $\notin$ NTIME$[2^{(1-\varepsilon)n}]$ [17]^{L}	refuted	A deterministic algorithm is a nondeterministic one ([A], after Corollary 5).
A10	NSETH for CNF with poly(m) clauses, or for poly-size circuits [55, Conj. 3]^{L}	survives	As A4.
A11	MA-SETH, AM-SETH	refuted (2016)	Williams [70]: MA proofs in c^n time, $c < 2$, for #SAT of sublinear-depth circuits; "strongly refute" both^{L}. For k-CNF also by (F1).
A12	Gap-ETH: no $2^{o(n)}$ (randomized) algorithm distinguishes satisfiable 3-CNFs from those not $(1-\varepsilon)$-satisfiable [31, 53], as quoted in [18]^{L}	refuted	An exact deterministic $2^{o(n)}$ decider distinguishes the cases. [52] also records Gap-ETH as false.
A13	ETH for Circuit-SAT of linear size; VSP-Circuit-SAT of size cn [25, Thm. 1.1(v)]^{L}	refuted	Tseitin gives $O(n)$ variables; gives algorithms.

	Hypothesis (source)	Status	Reason
A14	Circuit-, NC- and BP-SETH [4]^{L}	survives	Superlinear-size representations.
A15	FPT $\neq$ M[1] (miniaturised CNF of size $k \log n$) [33, §3.5]. Chen–Grohe: M[1] = FPT iff 3-SAT $\in$ DTIME($2^{o^{\mathrm{eff}}(n)}$) [21]^{L}	refuted	The iff needs an effective little-o; [A] (rate with absolute C) and [O] at $k = 3$ supply one. Gives algorithms for M[1]-complete problems.
A16	FPT $\neq$ W[1]	survives	ETH implies it, not conversely; [21, §6] list "ETH $\Leftrightarrow$ FPT $\neq$ W[1]?" as open^{L}.
A17	Low-dimension OVC: $\forall \varepsilon\, \exists c$, OV with $d = c \log n$ $\notin n^{2-\varepsilon}$ ("LDOVC" of [35])^{L}	survives	SETH support gone (it was via sparsification [13, Lemma 2.1]); [A] notes it does not refute OV.
A18	Moderate-dimension OVC: no $n^{2-\varepsilon}\,\mathrm{poly}(d)$ [35]^{L}	survives	Implied by CNF-SETH with $d = m$ [2, Lemma 1]; premise intact.
A19	k-OV, low and moderate dimension^{I}	survives	As A17, A18.
A20	Hitting Set conjecture [6]^{L}	survives	Not known to follow from SETH; implies OVC.
A21	Set Cover Conjecture: $\forall \varepsilon\, \exists k$, Set Cover with sets of size $\leq k \notin 2^{\varepsilon n}\,\mathrm{poly}(m)$, n the universe [25, Thm. 1.2]^{L}	survives	Not implied by SETH even before 2026; the natural witness has $\Theta(n \log n)$ bits, so (F2) does not apply.
A22	k-Hitting Set, k-Set Splitting, k-NAE-SAT need $2^{(1-\varepsilon)n}$ for large k, each equivalent to SETH [25, Thm. 1.1]^{L}	refuted	Equivalent to SETH; directly, they are k-CNFs on the universe. Gives algorithms.
A23	3SUM conjecture^{F}	survives	No SAT premise; see B26.
A24	APSP; combinatorial BMM; OMv^{F}	survives	Unrelated to SAT.
A25	k-Clique in $n^{\omega k/3-\varepsilon}$; min-, exact- and zero-weight k-Clique^{F}	survives	Unrelated.
A26	(ℓ, k)-Hyperclique hypothesis [47, Hyp. 1.4]^{L}	survives	Its refutation would give dense Max-k-SAT in $2^{(1-\varepsilon)n}$, which A27 leaves open; no converse from SAT.
A27	Dense Max-3-SAT $\notin 2^{(1-\varepsilon)n}$; Max-2-SAT, Max-Cut not below $2^{\omega n/3}$ (algorithm: [71])^{F}	survives	A counter over $\Theta(n^2)$ clauses has $\Theta(n^2)$ variables.
A28	Subset Sum $\notin 2^{(1/2-\varepsilon)n\,\mathrm{F}}$	survives	Weights are not local.
A29	Basic QSETH (CNF-SAT) [14]; its k-CNF variant^{L}	mixed	CNF form survives (as A4); k-CNF variant refuted [67, Table 2, row 23].
A30	PIH [51]^{L}	survives	Unproved again: both proofs, from Gap-ETH and from ETH [36], lose their premises.
A31	ETH for End-of-a-Line [60]^{L}	survives	A different, query-type hypothesis.
A32	Feige's random 3-SAT hypothesis [34]^{L}	survives	About polynomial-time average-case refutation.
A33	Almost-ETH: $\forall \gamma > 0$, 3-SAT $\notin O(2^{n^{1-\gamma}})$ [52]^{L}	survives	Both engines give $2^{n/(\log n)^{\Theta(1)}} = 2^{n^{1-o(1)}}$, not $2^{n^{1-\gamma}}$.

Remarks. Row A11 was refuted before 2026 and is listed for completeness. Row A15 is the one place where an *effective* rate matters. Chen and Grohe's miniaturisation isomorphism maps SUBEPT to FPT, and the explicit rates of both manuscripts make the conclusion M[1] = FPT unconditional modulo the refutations. Manurangsi proposed row A33 in the 27 October 2026 revision of [52], because of the refutations; its comment states that the earlier versions' ETH and Gap-ETH premises "are now known to be false".

4. Conditional theorems: four contradicted, one fires

In this section we classify the conditional theorems (Table 2) and then treat the one row whose premise is the negation of ETH, the circuit lower bound of Williams (Section 4.1).

Table 2: Conditional theorems "$H \Rightarrow X$", assuming the claimed refutations.

	Theorem (source)	Status	Reason
Parameterised and exact exponential			
B1	Treewidth DP optimality: no $(2-\varepsilon)^{\mathrm{tw}}$ IS, $(3-\varepsilon)^{\mathrm{tw}}$ DomSet, $(2-\varepsilon)^{\mathrm{tw}}$ Max-Cut, $(3-\varepsilon)^{\mathrm{tw}}$ OCT, $(q-\varepsilon)^{\mathrm{tw}}$ q-Col, $(2-\varepsilon)^{\mathrm{tw}}$ Partition into Triangles [48]^{L}	survives	The premise as proved is CNF-SETH (A4), with $m^{O(1)}$ clauses; no sparsification is used. No faster DP follows.

	Theorem (source)	Status	Reason
B2	Cut&Count pathwidth bounds: no $(3-\varepsilon)^{\mathrm{pw}}$ CVC, FVS, Steiner Tree; no $(4-\varepsilon)^{\mathrm{pw}}$ CDS, CFVS, COCT; Exact k-Leaf Spanning Tree [27, Thm. 1.3, App. D]^{L}	survives	Every reduction of App. D starts from CNF-SAT with n variables and m clauses: CNF-SETH. (Their Thm. 1.4 is in B21.)
B3	Pătraşcu–Williams: k-DomSet in $O(n^{k-\varepsilon})$ for some $k \geq 3 \Rightarrow$ CNF-SAT in $2^{\delta n}\,\mathrm{poly}(m)$, $\delta < 1$ [58, Thm. 2.1]^{L}	survives	CNF form.
B4	Pătraşcu–Williams: d-SUM on N numbers of $O(d \log N)$ bits, $d < N^{0.99}$, in $N^{o(d)} \Rightarrow k$-SAT in $2^{o(n)}$ [58, §5]^{L}	premise refuted	Conclusion of the contrapositive is now true; d-SUM bound open.
B13	ETH $\Rightarrow$ no $2^{o(n)}$ IS, Clique, VC, k-Col, DomSet on n vertices, any density [41, 49]^{L}	conclusion refuted	[A], Corollary 7: deterministic $O(2^{\delta n})$ for every δ, any number of edges, with counts by size and of k-colourings.
B14	ETH $\Rightarrow$ k-VC $\notin 2^{o(k)}\,\mathrm{poly}(n)$ (from B13, as $k \leq n$; cf. [15])^{L}	conclusion refuted^{I}	$2k$-vertex LP kernel [26, §2.5], then Corollary 7 (Section 5). Cai–Juedes: Max-SNP-complete problems have $2^{o(k)}p(n)$ algorithms iff 3-SAT $\in$ DTIME$(2^{o(n)})$.
B15	ETH $\Rightarrow$ no $2^{o(n)}$ Max-Cut, Max-k-SAT with $O(n)$ constraints^{F}	conclusion refuted^{I}	Unweighted, or $O(1)$-bit weights; Section 5. Polynomial weights: under $(\mathrm{H_A})$, or by the white-box extension of [O] to $\mathbb{Q}$-algebra weights in [8] (not a consequence of the top-level theorem of [O]). Dense: A27.
B16	ETH $\Rightarrow$ no $2^{o(n)}$ HC, HP, TSP, Steiner, FVS, CVC, CDS, Max-Leaf on sparse graphs ([49, Thm. 3.3] for Hamiltonian Path; folklore for the rest)	conclusion refuted under $(\mathrm{H_A})^{\mathrm{O}}$	HC, HP, TSP and their directed forms: machine-checked [63] (Section 5). Steiner, FVS, CVC, CDS, Max-Leaf: conjectured in our notes [68], not claimed. On dense graphs only the premise is refuted.
B19	ETH $\Rightarrow$ k-Clique $\notin f(k)n^{o(k)}$ [19, Thm. 5.5]^{L}	premise refuted	$2^{\delta n}$ does not contradict it at small k.
B20	ETH $\Rightarrow$ binary CSP $\notin f(G)n^{o(k/\log k)}$, k the treewidth [54]; #ETH $\Rightarrow$ #k-matching (bipartite) $\notin f(k)n^{o(k/\log k)}$ [24]^{L}	premise refuted	Parameter $k \ll n$. On a class of bounded-degree graphs of unbounded treewidth, Nguyen [56] decides binary CSP in $f(G)n^{O(k/\sqrt{\log k})}$ via the network engine of [A], contradicting the literal (unconditional) form of Marx's Conjecture 1.4 ($n^{o(k)}$); the exponent $o(k/\log k)$ of the theorem is not reached.
B21	ETH $\Rightarrow$ no $2^{o(d \log d)}$ Closest String, Distortion, Disjoint Paths by tw, ... [50]; no $2^{o(p \log p)}$ Cycle Packing [27, Thm. 1.4]^{L}	premise refuted	
B22	ETH $\Rightarrow$ $2^{\Omega(\sqrt{n})}$ planar bounds [49]; $2^{\Omega(n^{1-1/d})}$ for fat-object graphs [29] and Euclidean TSP [28]^{L}	premise refuted	The refutations give only $2^{n/\mathrm{polylog}\,n}$, worse than the upper bounds. Under Almost-ETH the reductions give $2^{n^{1/2-\gamma}}$-type bounds^{I}.
B11	Under the Set Cover Conjecture: Steiner Tree $\notin 2^{\varepsilon t}$ poly, CVC $\notin 2^{\varepsilon t}$ poly, Set Partitioning $\notin 2^{\varepsilon n}\,\mathrm{poly}(m)$, Subset Sum $\notin t^{\varepsilon}\,\mathrm{poly}(n)$ [25, Thm. 1.2]^{L}	survives	A21 intact.
B12	Subset Sum $\notin T^{1-\varepsilon}2^{o(n)}$ [3]^{L}	premise refuted	Reduces from k-SAT; the $T^{1-\varepsilon}\,\mathrm{poly}(n)$ form survives via B11.
Approximation			
B23	Gap-ETH $\Rightarrow$ FPT-inapproximability of k-Clique, k-DomSet, Biclique, induced matching, DkS [18]^{L}	premise refuted	Partly re-proved under W[1] $\neq$ FPT (survives): constant-factor k-Clique [46], $k^{1/H(k)}$ for k-Clique [44], any $F(k)$ for k-DomSet [45]. The ETH parts (KLM's $T(k)n^{o(k)}$; the $k^{o(1)}$ bounds discussed in [20]) lose their premise. No W[1] re-proof found for the others.
B24	ETH-hardness of approximation: 2-CSP to $\lvert V\rvert^{1-o(1)}$, Directed Steiner Network to $k^{1/4-o(1)}$ [32]; DkS [52]; best Nash [10]; free games [1]^{L}	mixed	DkS to $2^{(\log n)^{1-\varepsilon}}$ re-proved under Almost-ETH in [52] (v3): survives. [32, 10, 1]: premise refuted; quasi-polynomial reductions plausibly give weaker bounds under Almost-ETH^{I}.
B25	ETH $\Rightarrow$ PIH [36]; near-optimal $f(k)n^{o(k)}$-type bounds for approximate k-Clique and CSP under ETH [37]^{L}	premise refuted	See A30.

	Theorem (source)	Status	Reason
Problems in P			
B5	Edit distance $\notin n^{2-\varepsilon}$ [7]; also from NC-SETH [4]^{L}	survives	Reduction from CNF-SAT with M clauses: CNF-SETH.
B6	LCS, DTW: Most-OV with threshold, $n^{2-\varepsilon}\,\mathrm{poly}(d) \Rightarrow k$-LCS $\notin n^{k-\varepsilon}$ [2, Thm. 1, Conj. 1]; binary LCS, edit distance with costs, DTW [13]^{L}	survives	[2, Lemma 1]: Conj. 1 follows from Max-CNF-SAT hardness with $\mathrm{poly}(m)$ clauses. [13] is stated under k-SETH but proved from OVH, which CNF-SETH implies.
B7	Fréchet distance $\notin n^{2-\varepsilon}$; no 1.001-approximation in $(nm)^{1-\delta}$ [12]^{L}	survives	Stated from SETH′: no $O^*((2-\delta)^N)$ CNF-SAT.
B8	Diameter: no $(3/2-\varepsilon)$-approximation in $O(m^{2-\varepsilon})$, undirected unweighted [59, Thm. 1]^{L}	survives	Stated with conclusion "CNF-SAT in $O^*((2-\delta)^n)$". Later k-OV-based tradeoffs not checked here.
B9	Dynamic problems under SETH (#SSR, SubConn, SCC, ...) [5]^{L}	survives^{I}	Stated via k-SAT with $O(n)$ clauses (sparsification), which is refuted; but the constructions use one node per clause, so the proofs run from CNF-SETH. Triangle, BMM and 3SUM-based bounds untouched.
B10	Results using $d = c\log n$ exactly, stated "under SETH"	premise refuted	They survive if restated under low-dimension OVC (A17).
B26	NSETH $\Rightarrow$ no deterministic fine-grained reduction from SAT to 3SUM, APSP [17]^{L}	premise refuted	The barrier is lifted; no reduction follows.
Counting, quantum, circuits			
B17	#ETH bounds for the Tutte polynomial, the permanent and #IS [30, 23, 9]^{L}	mixed	Conclusion refuted: #2-SAT and #IS on n variables/vertices at any density ([A], Corollaries 6, 7); sparse permanent, #PM, Boolean #CSP, the integer hyperbolas H_q (local encodings; also, independently and earlier, [8]); $T(x,1)$ on sparse graphs under ($\mathrm{H_A}$) (conjectured, not claimed)^{O}. Dense permanent and the rest: premise refuted.
B18	Quantum-supremacy statements from ETH/SETH [55] and others	mixed	Of 24 rows in [67, Table 2]: 5 refuted, 5 premise refuted, 1 moot, 1 open, 12 survive.
B27	ETH false $\Rightarrow \mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(O(n))$ [72, Thm. 6.1]; see [43]^{L}	fires	Section 4.1.

Rows are grouped by area rather than by number; the numbering follows our working map in [68]. Pătraşcu and Williams state row B3 as "Hypothesis 1 implies that CNF-SAT has an improved algorithm". Section 2 of [58] remarks that the earlier $f(k)n^{o(k)}$ bound for k-DomSet of Chen et al. concerns only k-SAT, which is precisely the distinction of Section 2.

4.1. **The circuit lower bound fires.** Williams [72, Theorem 6.1] proved that if ETH is false then E^{NP} has no circuits of linear size. [A] draws this consequence and formalises it in Lean. Modulo the refutations, $\mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(O(n))$ is therefore unconditional. In a companion note [64] we track the constants through Williams' argument and the local reductions of Jahanjou, Miles and Viola [43]. Assuming Anthropic's #3-SAT theorem, in the Turing-machine form of the clause-count bound of Imported Theorem 1.1 that its Lean development provides, there are $\varepsilon > 0$ and $L_0 \in \mathrm{E}^{\mathrm{NP}}$ with no circuits of size $\varepsilon n\sqrt{\log n}$ for infinitely many n. We prove this in Lean on top of Anthropic's own circuit-lower-bound module (`enp_not_size_n_sqrt_log`; standard axioms; fresh kernel replay). It holds only infinitely often, and ε is not explicit. It gives nothing at $n\log n$ and nothing for NEXP. Carmosino et al. [17] note that disproving NSETH gives circuit lower bounds; for the k-CNF form refuted here this is subsumed by the above.

5. The new algorithms are few

The refutations give few explicit algorithms beyond k-SAT itself. In this section we list those we know of. We separate what needs only the formalised top-level statements from what needs internal results of [A]. Bhattacharjee [8] claims further weighted consequences, which we do not list.

5.1. **From the main theorems alone.**

(i) k-SAT, #k-SAT, $\oplus k$-SAT and k-TAUT; #CNF-SAT of any width with $O(n)$ clauses (either engine), and with $n2^{o(L(n))}$ clauses ([A]; Proposition 2.1); linear-size Circuit-SAT.

(ii) [A], Corollary 7: for every $\delta > 0$, deterministic $O(2^{\delta n})$ algorithms that count the independent sets, cliques, vertex covers and dominating sets of each size s, and the proper k-colourings, of a graph on n vertices with any number of edges. Decision for IS, Clique, VC and k-Colouring was implied before by [41]; Dominating Set and the counts are new in [A].

(iii) k-Hitting Set, k-Set Splitting, k-NAE-SAT, #2-SAT, sparse permanent, #PM, Boolean #CSP, and the integer Potts and flow hyperbolas of the Tutte plane on sparse graphs (local encodings, (F2)).

(iv) *Vertex Cover in* $2^{o(k)}+\mathrm{poly}(n)$ (our observation). The LP kernel leaves at most $2k$ vertices in polynomial time [26, §2.5]. Corollary 7 then counts vertex covers of size k in $2^{2\delta k}$ for every δ. A single algorithm in time $2^{O(k \log\log k/\sqrt{\log k})} + \mathrm{poly}(n)$ follows by running the explicit n-bound of Imported Theorem 1.1 on the $O(k)$-variable 3-CNF of the proof of Corollary 7. This contradicts the conclusion of Cai–Juedes' ETH bound (B14), consistently with their own equivalence.

(v) *Sparse Max-Cut and Max-k-SAT, unweighted* (our observation). Take $m = O(n)$ constraints with $O(1)$-bit weights. Encode each constraint by a forced indicator variable, add an $O(m)$-gate adder and a comparison with a target t, and apply Tseitin. The result is a 3-CNF with $O(n)$ variables whose models are the assignments of value exactly t. Corollary 6 counts them for each t, giving the exact optimum and the number of solutions of each value in $2^{\delta n}$. Weights up to W cost $O(m \log W)$ gates, so the main theorems do not cover polynomial weights. Bhattacharjee [8] obtains them by running the algorithm of [O] itself over truncated polynomial rings, a white-box extension.

(vi) M[1]-complete problems in FPT time (A15).

5.2. **From internal results of [A].** Our other results use three internal statements of [A] (a network-to-clique reduction, an algorithm for cliques of all levels, and the existence of its "pieces"), called ($\mathrm{H_A}$) in [63, 67]. They are not consequences of the top-level theorems. They are, however, theorems of [A]'s Lean development, and our Lean proofs are built on that development. So the results below that we call machine-checked use only the standard axioms (`propext`, `Classical.choice`, `Quot.sound`), have passed a fresh kernel replay, and stand or fall with [A]'s development.

(vii) *Sparse Hamiltonicity* [63]: the number of Hamiltonian cycles of a graph with n vertices and $O(n)$ edges in $O(2^{\delta n})$ for every δ, and in $2^{O(\max(m,n)/\sqrt{\log n})}$ in general; hence decision, search, Hamiltonian paths, sparse TSP with polynomial weights, directed cycles and ATSP. All are machine-checked in Lean, end to end on top of [A]'s development.

(viii) *Eulerian circuits* [65], a consequence of (vii). Counting Eulerian circuits is #P-complete [11]. For every $\delta > 0$ and Δ there is a machine that counts them on graphs of maximum degree at most Δ in time $(n+2)^C 2^{\delta m}$, and one machine does so for all graphs in $(n+2)^C 2^{Cm \log_2(\Delta+2)/\sqrt{\log_2(m+2)}}$. The reduction is Kotzig's bijection with the Hamiltonian cycles of the line graph of the double subdivision. Machine-checked; the formal object is Kotzig's transition-system definition.

(ix) *Exact quantum amplitudes* [67]: amplitudes of Clifford+T circuits with G two-qubit gates in $2^{O(G/\sqrt{\log G})}$, machine-checked. The polynomial-space variant and the marginal and sampling corollaries have written proofs only and are not claimed.

(x) *Deterministic factoring.* Rudich and Rousseau [61] used the counting engine to prove in Lean that one deterministic Turing machine computes the least prime factor of an n-bit integer within $C_k 2^{n/k}$ steps for every k, and conjectured the rate $2^{O(n/\sqrt{\log n})}$. Before 2026 every proven deterministic bound had a fixed positive exponent, the best being $N^{1/5+o(1)}$ [38]. The rate $2^{O(n/\sqrt{\log n})}$ was proved in Lean in a public repository [42] and on paper by Shunia [62], conditionally on two unrefereed preprints. We give an independent second Lean proof, along a different route [66]: one Turing machine outputs the least prime factor of N within $\lfloor C \cdot 2^{Cn/\sqrt{\log_2(n+2)}} \rfloor$ steps, by evaluating a two-dimensional Schönhage–Strassen multiplication network with Anthropic's counting theorem. Machine-checked.

(xi) *Further sparse problems, conjectured and not claimed* [68]: Steiner Tree, FVS, Connected VC and DS, Max-Leaf; the line $T(x,1)$ of the Tutte plane (forests counted by size) on sparse graphs, which is the line Brand, Dell and Roth [9] show #ETH-hard; sparse Max-Cut with polynomial weights. All are conditional on ($\mathrm{H_A}$). For the Tutte and connectivity problems the identities are checked numerically only; none is formalised. For Max-Cut and Max-k-SAT with polynomial weights, Bhattacharjee [8]

gives an earlier, independent route. It replaces (H_A) by the claim that the steps of [O] remain valid over a commutative $\mathbb{Q}$-algebra (unrefereed, not formalised).

5.3. **What gets no algorithm.** Nothing parameterised by treewidth or pathwidth gets an algorithm, except binary CSP on one class of bounded-degree graphs, with exponent $O(\mathrm{tw}/\sqrt{\log \mathrm{tw}})$ [56] (B20). Nor does any problem in P (edit distance, LCS, Fréchet, diameter, OV, 3SUM, APSP, k-Clique), Set Cover by universe size, Subset Sum, dense Max-CSP, dense Hamiltonicity, or a planar problem (where $2^{O(\sqrt{n})}$ is already better).

6. Reading the map by area

In this section we read the two tables area by area: exact and parameterised algorithms, approximation, problems in P, and counting and quantum.

6.1. **Exact and parameterised algorithms.** The treewidth and pathwidth bounds (B1, B2) are the clearest case of a premise that survives because of how it is stated; this is the first half of Example 1.3. Lokshtanov, Marx and Saurabh build, from a CNF formula with n variables and m clauses, a graph of size polynomial in $n+m$ and treewidth roughly $n/\log_2 c$ (the precise form varies by problem). So a $(c-\varepsilon)^{\mathrm{tw}}$ algorithm would decide CNF-SAT in time $2^{(1-\delta)n}\,\mathrm{poly}(m)$. Nothing in the reduction requires bounded width or few clauses, and Remark 2.2 shows that the new algorithms reach only $m \le n2^{o(L(n))}$. To refute B1 one would need CNF-SAT with, say, n^2 clauses of width up to n in time $2^{(1-\delta)n}$. The width reduction would then need $k \approx \log n$, where the bound of Imported Theorem 1.1 is $2^{\Theta(n(\log n)^{3/2}\log\log n)}$, far above 2^n. The other half of Example 1.3, row B13, is contradicted outright. Corollary 7 of [A] is a reduction in the other direction: it encodes Independent Set on n vertices, at any density, as a 3-CNF with $O(n)$ variables.

The ETH-based bounds for parameterised problems (B19–B22) lose their premise, but are not contradicted: a $2^{\delta n}$ algorithm says nothing about $f(k)n^{o(k)}$ when $k \ll n$. The two exceptions are structural. Chen and Grohe's equivalence (A15) is an "if and only if", so its other direction now yields M[1] = FPT. And Vertex Cover parameterised by solution size has a linear kernel, so the parameter and the number of vertices coincide after preprocessing, and the $2^{\delta n}$ algorithm of [A] becomes a $2^{o(k)}$ algorithm (B14). Whether W[1] $\neq$ FPT (A16) is unaffected: Chen and Grohe list its equivalence with ETH as open, and the refutations settle ETH without touching it.

6.2. **Approximation.** Gap-ETH is refuted by any exact $2^{o(n)}$ algorithm, so it falls with ETH; the gap does not help. The FPT-inapproximability results of Chalermsook et al. (B23) therefore lose their premise. So do the ETH-hardness results of Dinur–Manurangsi, Braverman–Ko–Weinstein and Aaronson–Impagliazzo–Moshkovitz (B24) and the ETH proof of PIH (B25). Two kinds of result survive. The first are the re-proofs under W[1] $\neq$ FPT: constant-factor k-Clique (Lin), $k^{1/H(k)}$-factor k-Clique (Karthik–Khot) and $F(k)$-factor k-DomSet (Karthik–Laekhanukit–Manurangsi). The second are results whose reductions are only mildly exponential, so that a weaker hypothesis suffices: Manurangsi's Densest-k-Subgraph bound, now stated under Almost-ETH (A33). The refutations give 3-SAT in $2^{n/(\log n)^c}$ for every constant $c < 1/2$. This is $2^{n^{1-o(1)}}$ but not $2^{n^{1-\gamma}}$, so Almost-ETH stands. We expect other quasi-polynomial-size reductions (Nash, free games, 2-CSP) to give weaker but non-trivial bounds under Almost-ETH; we have not checked this ([I]).

6.3. **Problems in P.** Here the map is almost entirely "survives". Edit distance, Fréchet distance and diameter are stated from CNF-SAT with poly(m) clauses (B5, B7, B8). LCS and DTW go through Most-OV or OV with dimension $d = m$ (B6), which CNF-SETH implies. What is lost is the derivation of the *low-dimension* OV conjecture ($d = c\log n$) from SETH, which uses sparsification to $O(n)$ clauses (A17, B10). OVC itself, 3SUM, APSP and the hyperclique hypothesis were never consequences of ETH. The dynamic lower bounds of Abboud and Vassilevska Williams (B9) are a borderline case. They are stated through sparsification, but by our reading proved by constructions that also run from CNF-SETH. The barrier of Carmosino et al. (B26), that NSETH forbids deterministic fine-grained reductions from SAT to 3SUM and APSP, disappears for k-SAT; no such reduction is known.

6.4. **Counting and quantum.** Counting statements behave like decision ones. #ETH and the k-CNF form of #SETH fall (A6, A7). Lower bounds proved by local reductions on sparse instances are contradicted by explicit algorithms (B17): Corollaries 6 and 7 of [A] count independent sets and models of 2-CNFs at any density, and local encodings handle sparse permanents, perfect matchings and the integer hyperbolas of the Tutte plane. The non-local cases (Hamiltonian cycles, forests) need (H_A). Dense permanents, and the Tutte

plane on dense graphs, are untouched. For the quantum statements (B18) the same split appears: sparse Clifford+T circuits fall under (H_A), while dense or average-case conjectures survive [67, §6].

7. Related work

We organise related work by idea: collections of ETH-based lower bounds, the structure of SETH and its variants, and the first responses to the refutations.

Collections of lower bounds. Lokshtanov, Marx and Saurabh [49] survey lower bounds based on ETH, and Chapter 14 of the textbook of Cygan et al. [26] treats lower bounds based on ETH. Both were written to *use* the hypotheses. In contrast, we take each bound as an object and ask which of its premises the refutations remove.

The structure of SETH.. Closest to our work is that of Cygan et al. [25], who map which problems are equivalent to SETH and which are tied to the Set Cover Conjecture instead. Calabro, Impagliazzo and Paturi [16] relate clause width to clause density, and Gao, Impagliazzo, Kolokolova and Williams [35] separate low- and moderate-dimension OV. Carmosino et al. [17] introduce the nondeterministic variants. These works draw the distinctions our map depends on. In contrast, they ask which hypotheses imply which; we ask what is left once the k-CNF forms are false. The width–density relation of [16] becomes the dividing line of Section 2.

Responses to the refutations. Manurangsi [52] restates his Densest-k-Subgraph bound under Almost-ETH. Bhattacharjee [8] extends the algorithm of [O] to weighted sums, Nguyen [56] decides binary CSP with a sublinear treewidth exponent on one class of graphs, and Rudich and Rousseau [61] factor in time $2^{o(n)}$. Our own notes count Hamiltonian cycles [63], Eulerian circuits [65] and quantum amplitudes [67], factor [66], and prove a superlinear circuit lower bound [64]; [67, Table 2] classifies the fine-grained quantum-supremacy conjectures, which row B18 summarises. These works add algorithms one problem at a time. In contrast, this note asks what the refutations do to the hypotheses and lower bounds themselves.

8. Summary: three findings

Table 3 gives the number of rows per status.

Status	Table A (33)	Table B (27)	Total
refuted	12	–	12
conclusion refuted	–	4 (B13–B16)	4
fires	–	1 (B27)	1
premise refuted	–	10	10
survives	20	9	29
mixed	1 (A29)	3 (B17, B18, B24)	4

Table 3. Counts per status. Refuted: A1–A3, A5–A7, A9, A11–A13, A15, A22. Premise refuted: B4, B10, B12, B19–B23, B25, B26. Survives: A4, A8, A10, A14, A16–A21, A23–A28, A30–A33, B1–B3, B5–B9, B11.

Three points stand out. First, the most-cited SETH lower bounds survive with an intact premise: those for treewidth DP and for edit distance, Fréchet distance, LCS and diameter. They are proved from CNF-SAT with polynomially many clauses. What dies is "k-SETH $\Rightarrow$ low-dimension OVC" and anything that genuinely needs sparsification. Second, Gap-ETH is refuted. So the Gap-ETH programme in parameterised inapproximability and the ETH proof of PIH lose their premises; the parts re-proved under W[1] $\neq$ FPT survive. Third, the frontier of the new algorithms in clause density is sharp in an unusual way: $n \cdot 2^{o(L(n))}$ clauses through the explicit rate of [A], constant density through [O], and nothing at $n^{1+\Omega(1)}$.

9. Limitations

- *The map stands or falls with the claimed refutations* [22, 57], whose expert review is pending. If they fail, every "refuted" becomes "open" and every "premise refuted" reverts to a valid conditional theorem; Proposition 2.1 and the observations of Section 5 lose their proofs. Results marked (H_A) use internal

statements of [A] that are not consequences of its top-level theorems. They are theorems of [A]'s Lean development, so these results stand or fall with that development.

- *A refutation removes premises.* Except through (F2) and the network theorems, it gives no algorithm, and "premise refuted" is never evidence that a conclusion is false.
- *The sources were checked, but the map is not exhaustive.* Every row was checked against its primary source; the nine rows that earlier rested on secondary sources were re-verified for this note (Section 1.5), and no row is now secondary. Rows marked ^{F} have no single primary statement. Rows marked ^{I} are our inferences and are argued only in the reason column. B8 omits later diameter tradeoffs. [A] reports more than 800 papers with results proved only under ETH or a hypothesis implying it.
- *Our written arguments are not machine-checked.* Proposition 2.1 and the observations (iv)–(v) of Section 5 are short written arguments, produced with AI assistance and not machine-checked or reviewed by a human expert. Only the results so labelled in Section 5 are machine-checked, on top of the labs' Lean developments. No human has yet checked that their formal statements match the printed ones.
- *Nothing is practical.* [A] notes that its bound does not beat 2^n for 3-CNFs with $5n$ clauses unless $n > 10^{10\,000}$. All algorithms here are galactic.

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